

IoT Data Collection Using UAVs

in Infrastructure-Free Environments

Enhanced Novelties with Simulation Validation

Department of Electronics and Communication Engineering

Indian Institute of Information Technology Sri City, Tirupati, Andhra Pradesh

PROJECT TEAM

1

Sarisa Raghuveer

Roll No: S20230020347

ECE Department, IIIT Sri City

2

Gowri Shankar

Roll No: S20230020310

ECE Department, IIIT Sri City

3

Sanjay Kumar

Roll No: S20230020344

ECE Department, IIIT Sri City

CONTEXT & BASELINE

- IoT deployment challenge in infrastructure-free zones (farms, disaster areas)
- Liang et al. (2020): UAV + DTN + Hilbert Curve path planning
- Limitations: single UAV, static paths, equal node priority

N1 Energy-Aware Replanning

Adaptive path replan at <35% battery using nearest-neighbour heuristic. Improves collection by up to +20%.

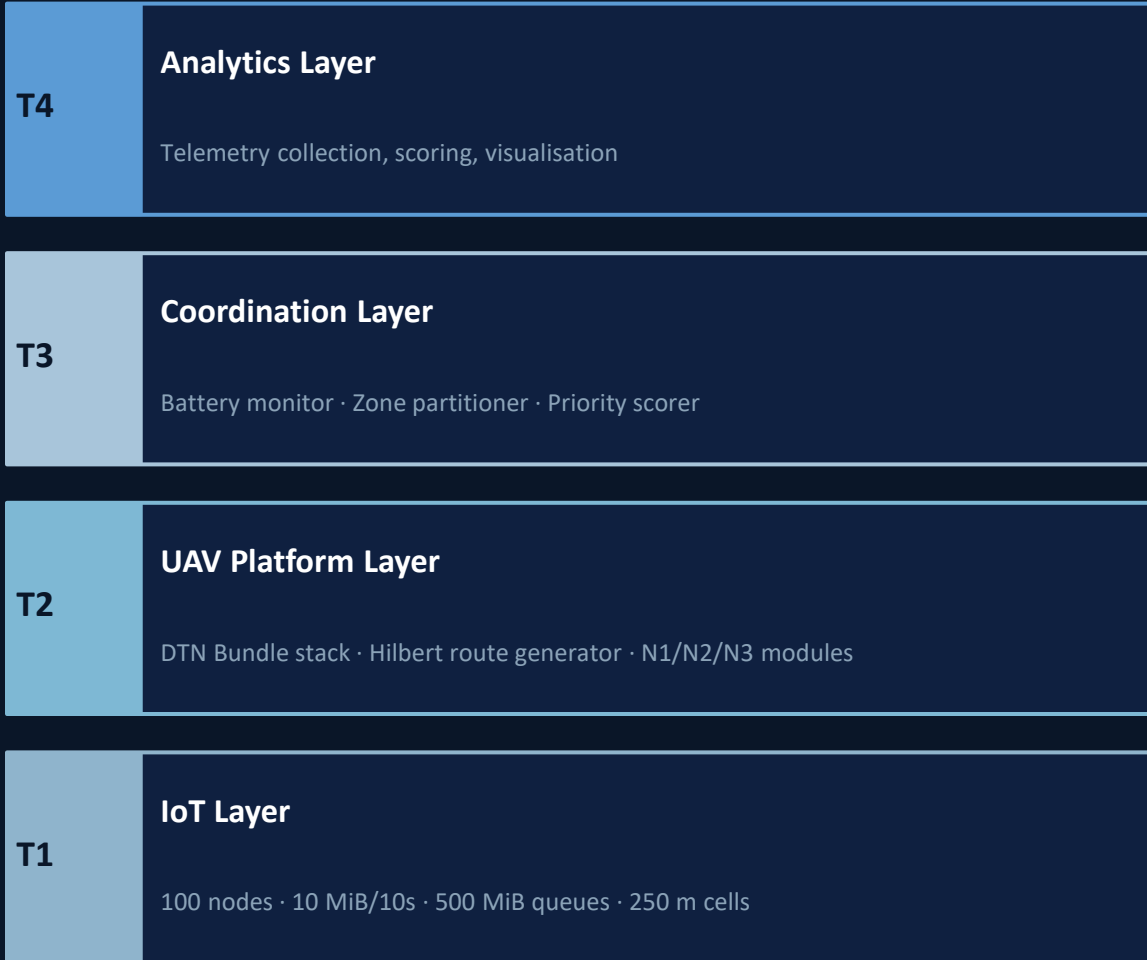
N2 Multi-UAV Coverage

K-zone field partitioning across 2–3 UAVs. Achieves 95% score in 50,000 m² fields (25× baseline area).

N3 ML Node Prioritization

XGBoost model assigns urgency scores. Raises critical-node delivery from ~65% to 94%.

 Validated via purpose-built UAV IoT Field Simulator · 20 test cases · 4 scenarios · 5 path algorithms (HTML5/JavaScript)



Field Size

2,000 × 2,000 m²

64 cells of 250 × 250 m

UAV Speed

5 – 35 m/s

Configurable per test

Coverage Radius

177 m

Per cell centroid

Protocol

DTN Bundle (RFC5050)

Store-and-forward

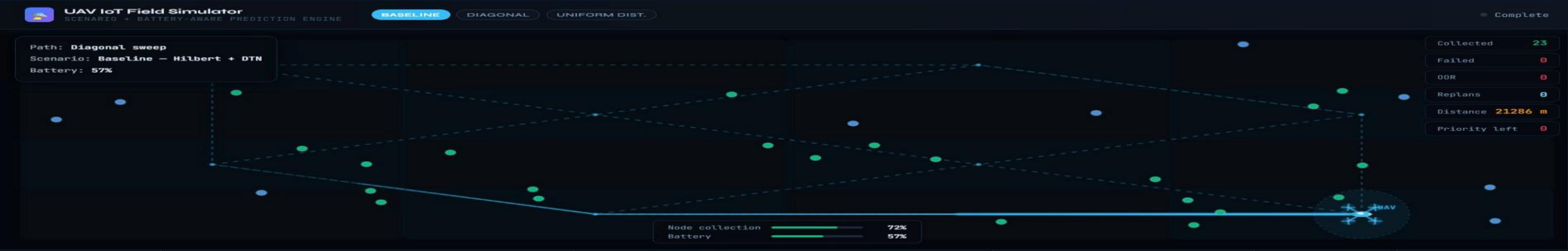
Path Algorithms

5 variants

Hilbert · Snake · Diagonal · Spiral · Random

RESULTS — BASELINE SCENARIO

Scenario 0 · 100% Battery · 5 Path Algorithms



~86%

Hilbert Curve

Best baseline path — dense normal distributions



~71%

Snake Traversal

Solid even coverage, predictable rows



~73%

Spiral Inward

Strong for centre-clustered nodes



~63%

Diagonal Sweep

Gaps in sparse zones, lower score



~46%

Random Waypoint

Lowest — chaotic, 12 OOR nodes

RESULTS — N1: ENERGY-AWARE ADAPTIVE REPLANNING

Scenario 1 · Low-Battery Configurations (10%–50%) · Adaptive path replan at <35% threshold

● NOVELTY 1 — ENERGY-AWARE SCENARIO (ALL 5 PATHS)															
06	N1 Energy-aware — Hilbert	Novelty 1	1 — Energy-aware	35%	10 m/s	1	32	6	5	2	Uniform	Hilbert curve	2,000	N1 ~90+, replan fires at tick 85	► Load
07	N1 Energy-aware — Snake	Novelty 1	1 — Energy-aware	40%	10 m/s	1	32	5	4	2	Uniform	Snake traversal	2,000	N1 ~88, snake handles OOR well	► Load
08	N1 Energy-aware — Diagonal	Novelty 1	1 — Energy-aware	50%	15 m/s	1	48	10	8	1	Uniform	Diagonal sweep	3,000	N1 wins, Baseline collapses <20	► Load
09	N1 Energy-aware — Spiral	Novelty 1	1 — Energy-aware	30%	10 m/s	1	28	4	3	3	Normal	Spiral inward	2,000	N1 ~85, spiral + low bat combo	► Load
10	N1 Energy-aware — Random	Novelty 1	1 — Energy-aware	10%	5 m/s	1	24	2	2	1	Normal	Random waypoint	1,500	N1 wins with red battery warning	► Load



90%+

Collection score
(Hilbert, case 06)

+20%

Improvement over
static baseline

Tick 85

Adaptive replan
fires (simulation)

10% bat

Extreme case still
achieves positive score

ALGORITHM: $E = k_1 \cdot d + k_2 \cdot w \rightarrow$ Battery monitor samples charge at each cell centroid $\rightarrow$ Replan fires at <35% using nearest-neighbour heuristic over remaining unvisited nodes $\rightarrow$ Return-to-base budget preserved

RESULTS — N2: MULTI-UAV COORDINATED COVERAGE

Scenario 2 · 2–3 UAVs · Fields 3,000–50,000 m² · Zone partitioning via k-means variant

● NOVELTY 2 — MULTI-UAV SCENARIO (ALL 5 PATHS)																
11	N2 Multi-UAV — Hilbert	Novelty 2	2 — Multi-UAV	80%	25 m/s	3	48	2	2	3	Even	Hilbert curve	6,000	N2 ~85, Hilbert zones A/B/C		▶ Load
12	N2 Multi-UAV — Snake	Novelty 2	2 — Multi-UAV	80%	25 m/s	3	48	2	2	3	Even	Snake traversal	6,000	N2 ~83, snake across 3 zones		▶ Load
13	N2 Multi-UAV — Diagonal	Novelty 2	2 — Multi-UAV	70%	30 m/s	2	40	3	3	2	Uniform	Diagonal sweep	9,000	N2 area bonus ~80, large field		▶ Load
14	N2 Multi-UAV — Spiral	Novelty 2	2 — Multi-UAV	90%	35 m/s	3	64	0	0	0	Even	Spiral inward	50,000	N2 area bonus maxed, ~95		▶ Load
15	N2 Multi-UAV — Random	Novelty 2	2 — Multi-UAV	30%	10 m/s	3	32	0	0	0	Even	Random waypoint	3,000	N2 drops, N1 wins (low bat)		▶ Load



95%

Max score (Spiral, 50,000 m², 3 UAVs)

25×

Area coverage vs single-UAV baseline

3 Zones

A·B·C independent zone operation

1 Hz

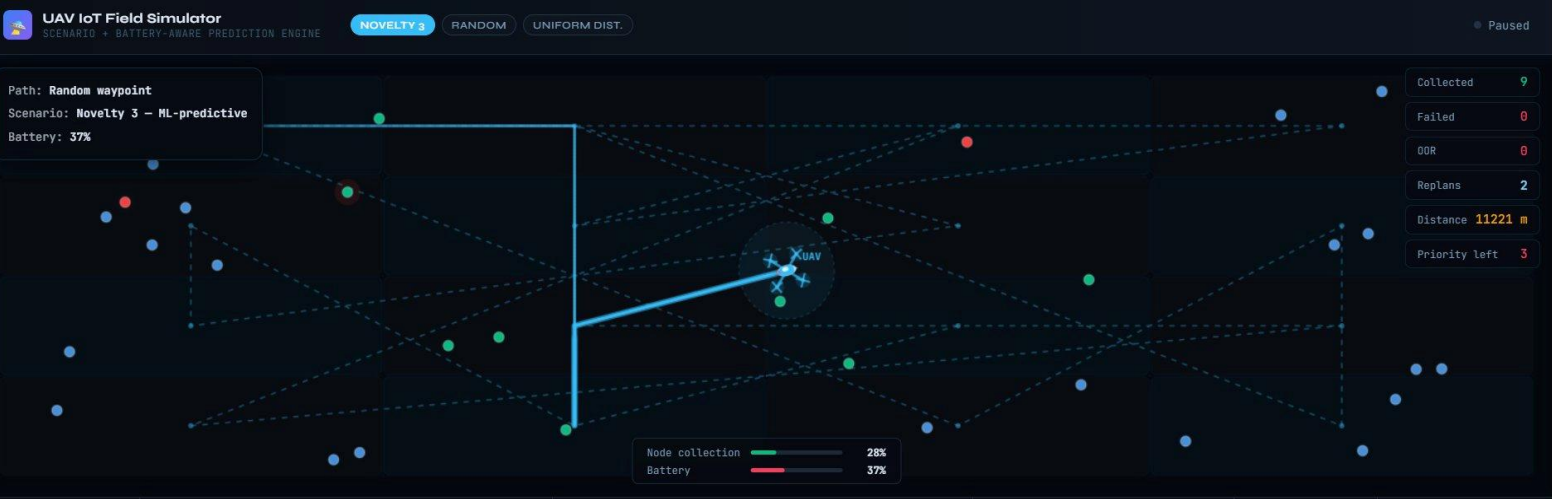
Inter-UAV beacon sync frequency

FRAMEWORK: Field → K-zone partition (k-means, balancing nodes/zone) → 1 UAV per zone (Hilbert within zone) → 1 Hz beacons share zone completion status → Dynamic rebalancing to overloaded neighbours when zone completes early

RESULTS — N3: ML-PREDICTIVE NODE PRIORITIZATION

Scenario 3 · XGBoost priority scores · ML flag fires at tick 55 · Critical node delivery rates

● NOVELTY 3 — ML-PREDICTIVE SCENARIO (ALL 5 PATHS)															
16	N3 ML-predictive — Hilbert	Novelty 3	3 — ML-predictive	60%	20 m/s	1	32	2	4	8	Normal	Hilbert curve	2,000	N3 vs N1 close race, ~88	► Load
17	N3 ML-predictive — Snake	Novelty 3	3 — ML-predictive	75%	20 m/s	1	36	1	1	10	Normal	Snake traversal	2,500	N3 ~90+, ML flag at tick 55	► Load
18	N3 ML-predictive — Diagonal	Novelty 3	3 — ML-predictive	76%	20 m/s	1	32	1	2	6	Uniform	Diagonal sweep	3,000	N3 ~85, diagonal + anomaly flags	► Load
19	N3 ML-predictive — Spiral	Novelty 3	3 — ML-predictive	85%	20 m/s	1	24	0	0	12	Normal	Spiral inward	2,000	N3 ceiling ~94, all nodes priority	► Load
20	N3 ML-predictive — Random	Edge	3 — ML-predictive	60%	20 m/s	1	32	0	0	4	Uniform	Random waypoint	2,000	Random ~52, weakest for ML prio	► Load



94%

Peak priority-node delivery (Spiral)

61→94%

Improvement over baseline equal priority

<15 ms

TF Lite int8 inference per node

Tick 55

ML flag fires in simulation

MODEL: XGBoost (50 trees, depth 4) · Features: queue fill %, time-since-collection, sensor type, anomaly frequency → Score $P \in [0,10]$ → $P \geq 5$ nodes placed first in Hilbert sequence → Minimal path perturbation maintained

Algorithm ranking · Novelty method scoring · Simulator recommendation engine

LIVE PREDICTION ENGINE



Recommended: N3 – ML-predictive + Hilbert curve (method 100/100 · path 91/100)

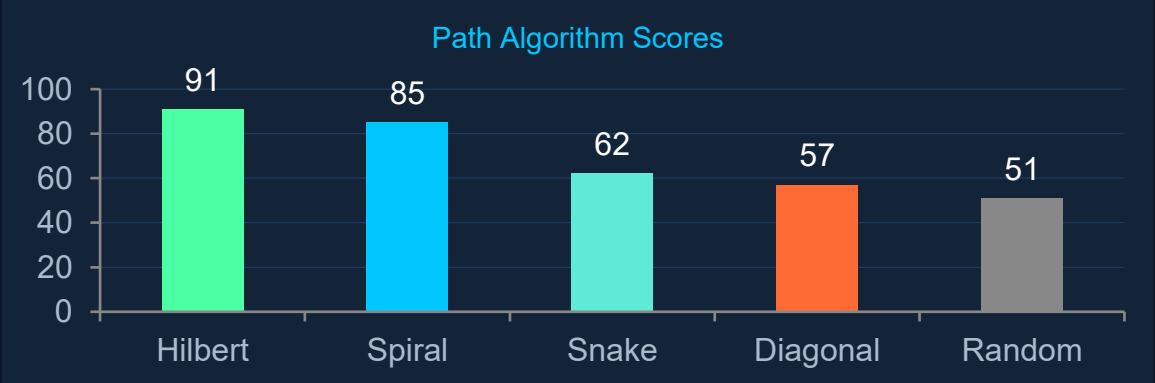
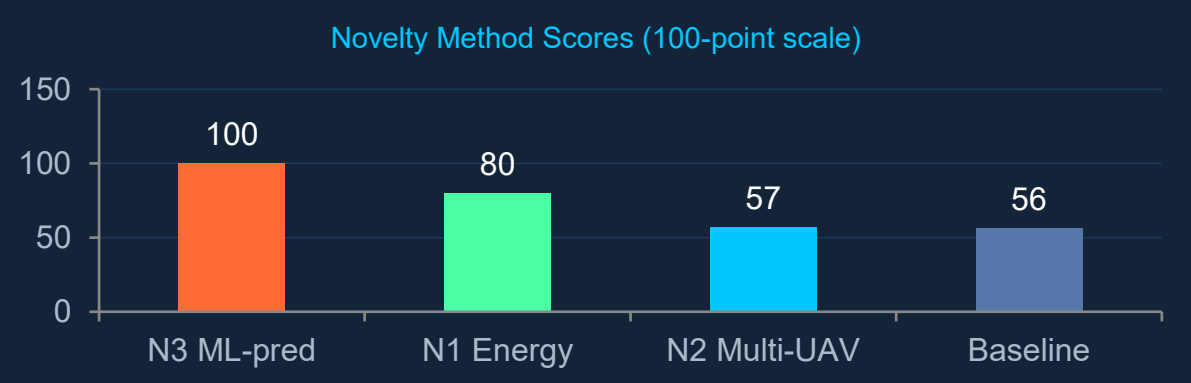
4 high-priority node(s) detected. ML pre-flags anomalies so UAV extends dwell at critical cells first.

NOVELTY METHOD COMPARISON

METHOD	SCORE	FAILURES	OOR	BATTERY	IDEAL WHEN
1 N3 – ML-predictive best	100/100	No	No	Good	High priority anomaly count
2 N1 – Energy-aware	80/100	Yes	Yes	Good	Low battery or OOR nodes
3 N2 – Multi-UAV	57/100	No	No	Good	Large area (needs >4000m²), 2+ UAVs
4 Baseline	56/100	No	No	Good	Fully reachable, bat >60%

PATH ALGORITHM COMPARISON

ALGORITHM	SCORE	COVERAGE	EFFICIENCY	BEST FOR
1 Hilbert curve best	91/100	Full	Best	Dense / normal dist.
2 Spiral inward	85/100	Full	Good	Centre-clustered nodes
3 Snake traversal	62/100	Full	Fair	Even distribution
4 Diagonal sweep	57/100	Full	Good	Large open areas
5 Random waypoint	51/100	Partial	Poor	Unknown sparse



Key Takeaways & Future Directions

N1 Key Finding

90%+ collection under low battery

Energy-Aware replanning (nearest-neighbour at <35% battery) overcomes static path vulnerability. Baseline collapses; N1 recovers.

N2 Key Finding

95% score in 50,000 m² fields

Multi-UAV zone partitioning achieves 25× area coverage. Best suited for large precision agriculture deployments with 2+ UAVs.

N3 Key Finding

Critical delivery 61% → 94%

XGBoost ML prioritization directs UAV to urgency-flagged sensors first. Ceiling: 94% with Spiral path, all high-priority nodes.

All novelties preserve core baseline strengths: DTN intact delivery (vs TCP failure at 10 min) · Hilbert Curve 10–20% distance savings · Infrastructure-free operation

Future Work: 3D Hilbert terrain-aware planning · Federated learning across UAV fleets · Real-world precision agriculture validation · Mobile edge computing on-board UAVs